# JEE Mathematics: Quick Review with Grading and Practice Map - Complex Number (Batch XII)

Grade shows how often the item is tested in the JM exercise of this batch, the JA exercise and the JEE Main Question Bank taken together (125 questions; a question that appears in two of these is counted once): High = 11 or more questions, Medium = 1 to 10 questions, Low = none. Low only means the item is not tested in these questions; it can still be important for understanding. Practice map lists the questions that test the item in the sheet exercises (J-I, J-II, J-M, J-A, C-I, C-II), the DPPs and the Question Bank (QB). 'Main concept' = the question is mainly about this item; 'Also used in' = the question needs it as a secondary step.

## Introduction to Complex Number

### Question 01

What is the real part and imaginary part of a complex number and how are they represented?

**Model Answer**

$z = x + iy$ $(x, y \in \mathbb{R},\ i = \sqrt{-1})$: real part $\mathrm{Re}(z) = x$, imaginary part $\mathrm{Im}(z) = y$ (a real number, without $i$).

Ex: $z = 3 - 4i$: $\mathrm{Re}(z) = 3$, $\mathrm{Im}(z) = -4$.

**Grade** - Medium

Practice map, main concept: none

Also used in: DPP: 1(11, 14) | Question Bank: 315, 341

### Question 02

What is a purely real / purely imaginary / imaginary number?

**Model Answer**

Purely real: $\mathrm{Im}(z) = 0$; purely imaginary: $\mathrm{Re}(z) = 0,\ \mathrm{Im}(z) \neq 0$; imaginary (non-real): $\mathrm{Im}(z) \neq 0$. 0 is purely real and also taken as purely imaginary.

Ex: 5 is purely real, $-2i$ is purely imaginary, $1 + i$ is imaginary.

**Grade** - Low

Practice map, main concept: none

Also used in: Sheet: C-I 1 | DPP: 1(1-2)

## Integral Powers of Iota

### Question 03

How to simplify a higher integral power of iota?

**Model Answer**

$i^2 = -1,\ i^3 = -i,\ i^4 = 1$; divide the power by 4: $i^{4k+r} = i^r$. Sum of four consecutive powers $i^n + i^{n+1} + i^{n+2} + i^{n+3} = 0$.

Ex: $i^{103} = i^3 = -i$; $\sum_{n=1}^{13}(i^n + i^{n+1}) = i - 1$.

**Grade** - Medium

Practice map, main concept: Sheet: J-I 6; J-II 6; J-M 7 | DPP: 1(12) | Question Bank: 329

Also used in: Sheet: J-M 18 | DPP: 6(9) | Question Bank: 355

## Argand Plane

### Question 04

Can you plot any given complex number on the complex plane?

**Model Answer**

$z = x + iy$ is the point $(x, y)$: real part on the $x$-axis (real axis), imaginary part on the $y$-axis (imaginary axis).

Ex: $-2 + 3i$ is the point $(-2, 3)$ in the second quadrant.

**Grade** - Low

Practice map, main concept: none

Also used in: none

### Question 05

Where is the position of a purely real / purely imaginary number on the complex plane?

**Model Answer**

Purely real numbers lie on the real $(x)$ axis; purely imaginary numbers lie on the imaginary $(y)$ axis.

Ex: $z = -7$ is $(-7, 0)$; $z = 4i$ is $(0, 4)$.

**Grade** - Low

Practice map, main concept: none

Also used in: none

## Algebra of Complex Number

### Question 06

How to add / subtract / multiply and divide two or more complex numbers?

**Model Answer**

$$(a + ib) \pm (c + id) = (a \pm c) + i(b \pm d), \qquad (a + ib)(c + id) = (ac - bd) + i(ad + bc)$$

Two complex numbers are equal iff real parts and imaginary parts are equal separately.

Ex: $(2 + 3i)(1 - 2i) = 2 - 4i + 3i + 6 = 8 - i$.

**Grade** - Medium

Practice map, main concept: Sheet: J-I 7; J-II 13, 18; J-M 26; C-II 5-7, 14, 17, 20 | DPP: 1(2, 4, 11, 15-16, 18); 2(4, 17) | Question Bank: 311, 334, 349, 353

Also used in: Sheet: J-II 9, 11; J-A 12; C-II 21 | DPP: 1(9, 17); 2(18); 5(14); 6(8) | Question Bank: 309

### Question 07

What to do when a complex number is given in the denominator?

**Model Answer**

Multiply numerator and denominator by the conjugate of the denominator:

$$\frac{a + ib}{c + id} = \frac{(a + ib)(c - id)}{c^2 + d^2}$$

Ex: $\dfrac{1 + 2i}{2 - i} = \dfrac{(1 + 2i)(2 + i)}{5} = \dfrac{5i}{5} = i.$

**Grade** - Medium

Practice map, main concept: DPP: 1(9); 2(18) | Question Bank: 309, 315

Also used in: Sheet: J-I 6, 12; J-M 9; J-A 24 | DPP: 1(10, 15, 18); 3(14)

**Question 08**

In which type of numbers is inequality defined and in which is it not defined?

**Model Answer**

Inequalities ($<, >$) are defined only for real numbers; $z_1 > z_2$ has no meaning for non-real numbers. Moduli, real and imaginary parts are real, so $|z_1| > |z_2|$, $\text{Re}(z) \geq 0$ are allowed.

Ex: $3 + 2i > 1 + i$ is meaningless, but $|3 + 2i| > |1 + i|$ is true ($\sqrt{13} > \sqrt{2}$).

**Grade** - Low

Practice map, main concept: none

Also used in: DPP: 2(8); 3(11)

**Question 09**

How to solve an equation in $z$ that also contains $\bar{z}$ or $|z|$ (e.g. $z^n = \bar{z}$, $z^2 + |z| = 0$)?

**Model Answer**

Either take modulus of both sides first (gives $|z|$), then use $z\bar{z} = |z|^2$ or arguments; or put $z = x+iy$ and equate real and imaginary parts. Do not forget $z = 0$.

Ex: $z^3 = \bar{z}$: $|z|^3 = |z| \Rightarrow |z| = 0$ or 1; for $|z| = 1$, $z^4 = z\bar{z} = 1$; so $z = 0, \pm 1, \pm i$: 5 solutions.

**Grade** - High

Practice map, main concept: Sheet: J-I 2, 16; J-II 5, 12; J-M 4, 24, 32, 46-47; J-A 18, 22, 25; C-I 14 | DPP: 1(1, 6); 3(16) | Question Bank: 323, 332, 337, 344

Also used in: Sheet: J-II 48; J-A 14 | DPP: 3(15) | Question Bank: 319

## Square Root of a Complex Number

**Question 10**

How to find the square root of a complex number?

**Model Answer**

Let $\sqrt{a + ib} = x + iy$: $x^2 - y^2 = a$, $2xy = b$, $x^2 + y^2 = \sqrt{a^2 + b^2}$. Then $x^2, y^2$ follow; the sign of $b$ fixes the signs of $x, y$ (same signs if $b > 0$).

Ex: $\sqrt{9 + 40i} = \pm(5 + 4i)$; $\sqrt{-11 - 60i} = \pm(5 - 6i)$.

**Grade** - Low

Practice map, main concept: DPP: 2(16)

Also used in: Sheet: J-II 1, 10; C-I 10 | DPP: 1(3)

**Question 11**

How to solve a quadratic equation if its coefficients are complex numbers?

**Model Answer**

Use $z = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ and find $\sqrt{b^2 - 4ac}$ as a complex square root. Roots need not be conjugates (that holds only for real coefficients).

Ex: $z^2 - (3 - 2i)z - (5i - 5) = 0$: $D = (3 - 2i)^2 + 4(5i - 5) = -15 + 8i = (1 + 4i)^2$, so $z = 2 + i$ or $1 - 3i$.

**Grade** - Medium

Practice map, main concept: Sheet: J-I 9; J-II 1, 10-11; C-I 10 | DPP: 1(3, 5); 3(4) | Question Bank: 320

Also used in: Sheet: J-A 1 | DPP: 3(6); 4(8)

## Conjugate and its Properties

### Question 12

How to find the conjugate of a complex number expressed as $a + ib$?

**Model Answer**

$\overline{a+ib} = a - ib$ (change the sign of the imaginary part). $z + \bar{z} = 2\,\mathrm{Re}(z)$, $z - \bar{z} = 2i\,\mathrm{Im}(z)$, $\bar{\bar{z}} = z$.

Ex: $\overline{-3+5i} = -3 - 5i$.

**Grade** - Low

Practice map, main concept: Sheet: J-II 8-9; C-II 21 | DPP: 1(14, 17)

Also used in: Sheet: C-II 14

### Question 13

What does the conjugate of a complex number represent on the complex plane?

**Model Answer**

$\bar{z}$ is the mirror image of $z$ in the real axis: $(x, y) \to (x, -y)$.

Ex: $z = 2 + 3i$ at $(2, 3)$, $\bar{z}$ at $(2, -3)$.

**Grade** - Low

Practice map, main concept: none

Also used in: DPP: 2(1)

### Question 14

How to prove / what to do when a given complex number is to be purely real?

**Model Answer**

$z$ is purely real $\iff z = \bar{z} \iff \mathrm{Im}(z) = 0$. For a quotient, equate it to its conjugate and cross-multiply (or make $\mathrm{Im} = 0$).

Ex: $z$ non-real and $\dfrac{1 - z + z^2}{1 + z + z^2}$ real $\Rightarrow z\bar{z} = 1$, i.e. $|z| = 1$.

**Grade** - Medium

Practice map, main concept: Sheet: J-I 24; J-II 2; J-M 12; J-A 1, 21, 24; C-I 1, 19 | DPP: 1(13); 3(5, 10) | Question Bank: 308, 339

Also used in: Sheet: J-II 36; J-M 40; C-II 32 | Question Bank: 312

### Question 15

How to prove / what to do when a given complex number is to be purely imaginary?

**Model Answer**

$z$ is purely imaginary $\iff z + \bar{z} = 0 \iff \mathrm{Re}(z) = 0$ (with $z \neq 0$).

Ex: $\dfrac{z-1}{z+1}$ purely imaginary $\Rightarrow \dfrac{z-1}{z+1} + \dfrac{\bar{z}-1}{\bar{z}+1} = 0 \Rightarrow z\bar{z} = 1$, $|z| = 1$.

**Grade** - Medium

Practice map, main concept: Sheet: J-I 12, 17; J-M 9, 14, 40; C-I 8 | Question Bank: 312, 338

Also used in: Sheet: J-I 26; J-II 37 | DPP: 1(13); 3(5); 5(5) | Question Bank: 330

**Question 16**
How to take the conjugate of a complex number expressed as a sum / product of several numbers?

**Model Answer**

$$\overline{z_1 \pm z_2} = \bar{z}_1 \pm \bar{z}_2, \quad \overline{z_1 z_2} = \bar{z}_1 \, \bar{z}_2, \quad \overline{\left(\frac{z_1}{z_2}\right)} = \frac{\bar{z}_1}{\bar{z}_2}, \quad \overline{z^n} = (\bar{z})^n$$

Ex: $z^5 + (\bar{z})^5 = z^5 + \overline{z^5} = 2\,\mathrm{Re}(z^5)$.

**Grade** - Medium
Practice map, main concept: Sheet: C-I 3
Also used in: Sheet: J-M 42

## Modulus and its Properties

**Question 17**
How to find the modulus of a complex number expressed as $a + ib$?

**Model Answer**

$$|a + ib| = \sqrt{a^2 + b^2} \ (\geq 0)$$

Ex: $|3 - 4i| = 5$.

**Grade** - Medium
Practice map, main concept: DPP: 2(8)
Also used in: Sheet: J-A 23; C-I 8 | DPP: 1(6)

**Question 18**
What does the modulus of a complex number represent on the complex plane?

**Model Answer**
$|z|$ = distance of the point $z$ from the origin.
Ex: all $z$ with $|z| = 2$ form the circle of radius 2 about $O$.

**Grade** - Low
Practice map, main concept: none
Also used in: none

**Question 19**
What is the relation between $z$, its conjugate and $|z|$?

**Model Answer**

$$z\,\bar{z} = |z|^2, \qquad |z| = |\bar{z}| = |-z|, \qquad |z| = 1 \iff \bar{z} = \frac{1}{z}$$

Ex: $z = 1 + 2i$: $z\bar{z} = 1 + 4 = 5 = |z|^2$.

**Grade** - Medium
Practice map, main concept: DPP: 2(1) | Question Bank: 316
Also used in: Sheet: J-I 31; J-II 5; J-A 25

**Question 20**
How to take the modulus of a complex number expressed as a product of several numbers?

**Model Answer**

$$|z_1 z_2 \cdots z_n| = |z_1||z_2| \cdots |z_n|, \qquad \left|\frac{z_1}{z_2}\right| = \frac{|z_1|}{|z_2|}, \qquad |z^n| = |z|^n$$

Ex: $\left|\dfrac{(1+i)^6}{(1-i\sqrt{3})^3}\right| = \dfrac{(\sqrt{2})^6}{2^3} = 1.$

**Grade** - Medium

Practice map, main concept: Sheet: J-I 3 | DPP: 3(2, 8); 6(8)
Also used in: Sheet: J-II 12; C-I 9; C-II 7 | DPP: 3(17) | Question Bank: 334

**Question 21**
What is the difference between the square of $z$ and the square of $|z|$?

**Model Answer**
$z^2 = a^2 - b^2 + 2iab$ is complex; $|z|^2 = a^2 + b^2 = z\bar{z}$ is real. $z^2 = |z|^2$ only when $z$ is real.
Ex: $z = 1 + i$: $z^2 = 2i$ but $|z|^2 = 2$.

**Grade** - Low

Practice map, main concept: none
Also used in: none

**Question 22**
What is the difference in finding the conjugate and the modulus of a complex number expressed as a sum of several numbers?

**Model Answer**
Conjugate distributes over a sum: $\overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2$; modulus does not: $|z_1 + z_2| \leq |z_1| + |z_2|$ (find the sum first, then its modulus).
Ex: $|1 + i| = \sqrt{2}$, but $|1| + |i| = 2$.

**Grade** - Low

Practice map, main concept: none
Also used in: none

**Question 23**
If two complex numbers are equal, then what is the relation between their moduli?

**Model Answer**
$z_1 = z_2 \Rightarrow |z_1| = |z_2|$ (the converse is false). Taking modulus of both sides of an equation is often the first step.
Ex: $|3 + 4i| = |5|$ but $3 + 4i \neq 5$.

**Grade** - Low

Practice map, main concept: none
Also used in: Sheet: J-I 2; C-I 14 | DPP: 1(5); 3(16)

## Argument and its Properties

**Question 24**

How to find the argument of a complex number expressed as $a + ib$?

**Model Answer**

Find $\alpha = \tan^{-1}\left|\frac{b}{a}\right|$ and use the quadrant: 1st: $\alpha$, 2nd: $\pi - \alpha$, 3rd: $-(\pi - \alpha)$, 4th: $-\alpha$ (principal value).

Ex: $\arg(-1 + i\sqrt{3}) = \pi - \frac{\pi}{3} = \frac{2\pi}{3}$; $\arg(-1 - i) = -\frac{3\pi}{4}$.

**Grade** - Medium

Practice map, main concept: Sheet: J-I 1 | DPP: 1(10)

Also used in: Sheet: J-I 29; J-M 21; J-A 15 | Question Bank: 340

**Question 25**

What is the difference between argument and amplitude (principal value of argument)?

**Model Answer**

Argument = any angle $\theta$ of $\vec{Oz}$ with the positive real axis (many values, differing by $2\pi$); amplitude / principal argument = the one value in $(-\pi, \pi]$.

Ex: $z = i$: arguments $\frac{\pi}{2} + 2k\pi$, principal argument $\frac{\pi}{2}$.

**Grade** - Medium

Practice map, main concept: Sheet: J-A 13 | DPP: 3(23)

Also used in: DPP: 2(5, 19-20)

**Question 26**

How to find the general value of the argument?

**Model Answer**

$$\mathrm{Arg}(z) = \arg(z) + 2k\pi, \quad k \in \mathbb{Z}$$

Ex: $z = 1 - i$: general argument $-\frac{\pi}{4} + 2k\pi$.

**Grade** - Low

Practice map, main concept: none

Also used in: Sheet: J-I 35

**Question 27**

What does the argument of a complex number represent on the complex plane?

**Model Answer**

The angle made by the line from $O$ to $z$ with the positive real axis (anticlockwise positive). $\arg(z) = \theta$ (fixed) is a ray from $O$ (origin excluded).

Ex: $\arg z = -\frac{\pi}{3}$: ray in the fourth quadrant making $60^\circ$ below the $x$-axis.

**Grade** - Low

Practice map, main concept: Sheet: J-II 20, 41, 45; C-I 18 | DPP: 2(5, 15)

Also used in: Sheet: C-I 13 | DPP: 2(14); 3(13)

**Question 28**
If two complex numbers are equal, then what is the relation between their arguments?

**Model Answer**
Equal principal arguments; general arguments differ by $2k\pi$. A complex number is fixed by modulus and argument together.
Ex: $2e^{i\pi/3} = re^{i\theta} \Rightarrow r = 2,\ \theta = \dfrac{\pi}{3} + 2k\pi.$

**Grade** - Low
Practice map, main concept: none
Also used in: Sheet: J-I 37

**Question 29**
What is the argument of complex numbers lying on the real and imaginary axes?

**Model Answer**
Positive real: 0; negative real: $\pi$; positive imaginary: $\dfrac{\pi}{2}$; negative imaginary: $-\dfrac{\pi}{2}$; $\arg(0)$ is not defined.
Ex: $\arg(-5) = \pi$, $\arg(-3i) = -\dfrac{\pi}{2}$.

**Grade** - Low
Practice map, main concept: none
Also used in: none

## Polar Form and Exponential Form

**Question 30**
What is the polar form? Explain all its components.

**Model Answer**

$$z = r(\cos\theta + i\sin\theta), \qquad r = |z|,\ \theta = \arg(z)$$

$r \geq 0$ is the modulus and $\theta$ the argument (any value; the principal one is usual).

**Grade** - Low
Practice map, main concept: none
Also used in: none

**Question 31**
How to express a complex number in polar form if it is given in $a + ib$ form?

**Model Answer**
Find $r = \sqrt{a^2 + b^2}$ and $\theta = \arg(a + ib)$ (by quadrant), then write $r(\cos\theta + i\sin\theta)$.
Ex: $-1 + i = \sqrt{2}\left(\cos\dfrac{3\pi}{4} + i\sin\dfrac{3\pi}{4}\right).$

**Grade** - Medium
Practice map, main concept: DPP: 4(1)
Also used in: Sheet: J-M 1, 11; C-I 16 | DPP: 2(12); 3(19) | Question Bank: 316, 354

**Question 32**
What does $\mathrm{CiS}(\theta)$ mean?

**Model Answer**

$$\mathrm{CiS}\,\theta = \cos\theta + i\sin\theta = e^{i\theta}, \qquad \mathrm{CiS}\,\alpha \cdot \mathrm{CiS}\,\beta = \mathrm{CiS}(\alpha+\beta)$$

Ex: $2\,\mathrm{CiS}\dfrac{\pi}{2} = 2i$.

**Grade** - Medium

Practice map, main concept: Sheet: J-A 27 | DPP: 2(7)

Also used in: none

**Question 33**
What is the exponential form? Explain all its components.

**Model Answer**
$z = re^{i\theta}$ with Euler's formula $e^{i\theta} = \cos\theta + i\sin\theta$; $r = |z|$, $\theta = \arg z$.
$\cos\theta = \dfrac{e^{i\theta}+e^{-i\theta}}{2}$, $\sin\theta = \dfrac{e^{i\theta}-e^{-i\theta}}{2i}$; $|e^{i\theta}| = 1$.

**Grade** - Medium

Practice map, main concept: Sheet: C-I 21 | DPP: 3(15) | Question Bank: 319

Also used in: Sheet: J-A 20 | DPP: 2(7); 5(1)

**Question 34**
How to express a complex number in exponential form if it is given in $a + ib$ form?

**Model Answer**
Find $r$ and $\theta$ exactly as for the polar form, then write $re^{i\theta}$.
Ex: $1 - i\sqrt{3} = 2e^{-i\pi/3}$.

**Grade** - Low

Practice map, main concept: none

Also used in: none

**Question 35**
When to use the polar form in questions?

**Model Answer**
When the question needs $\cos, \sin$ of the argument explicitly: real / imaginary parts of products and powers, comparing with $a + ib$, trigonometric conditions.
Ex: $(1+i)^4 = (\sqrt{2})^4(\cos\pi + i\sin\pi) = -4$.

**Grade** - Low

Practice map, main concept: none

Also used in: none

**Question 36**
When to use the exponential form in questions?

**Model Answer**
For products, quotients, powers and roots (arguments add / subtract / multiply), for $|z| = 1$ ($z = e^{i\theta}$) and for rotation.
Ex: $\dfrac{(1+i)^8}{(1-i)^6} = \dfrac{16e^{2\pi i}}{8e^{-3\pi i/2}} = 2e^{i7\pi/2} = -2i$.

**Grade** - Medium

Practice map, main concept: Sheet: J-I 26; J-II 31; J-A 23; C-II 2

Also used in: none

**Question 37**

When will we prefer the polar form, and when the exponential form, over the other form?

**Model Answer**

Exponential: algebra of many factors, powers, roots, $z = e^{i\theta}$ substitutions, rotation. Polar: final answer in cos / sin, or when real and imaginary parts must be compared.

Ex: $|z| = 1 \Rightarrow \dfrac{z}{1+z^2} = \dfrac{1}{e^{-i\theta}+e^{i\theta}} = \dfrac{1}{2\cos\theta}$ (real).

**Grade** - Low

Practice map, main concept: none

Also used in: none

**Question 38**

How to find the argument of a complex number expressed as a product / quotient of several numbers?

**Model Answer**

$$\arg(z_1 z_2) = \arg z_1 + \arg z_2 + 2k\pi, \qquad \arg\left(\frac{z_1}{z_2}\right) = \arg z_1 - \arg z_2 + 2k\pi$$

Choose $k$ so that the result lies in $(-\pi, \pi]$. Ex: $\arg\left((1+i\sqrt{3})(1+i)\right) = \dfrac{\pi}{3} + \dfrac{\pi}{4} = \dfrac{7\pi}{12}$.

**Grade** - Medium

Practice map, main concept: DPP: 2(13); 3(17, 19); 4(3); 5(4); 6(2) | Question Bank: 355

Also used in: DPP: 3(8); 4(9)

**Question 39**

How to find the argument of the conjugate of a complex number?

**Model Answer**

$\arg(\bar{z}) = -\arg(z)$ ($z$ not negative real); $\arg(-z) = \arg z \pm \pi$.

Ex: $\arg(\overline{1+i}) = -\dfrac{\pi}{4}$.

**Grade** - Medium

Practice map, main concept: none

Also used in: Sheet: J-I 4-5; J-M 1, 22 | DPP: 1(7)

**Question 40**

What is the pattern of the properties of argument?

**Model Answer**

Argument behaves like a logarithm: product → sum, quotient → difference, power → multiple, conjugate → negative (each up to $2k\pi$).

Ex: $|z| = |w|$, $\arg z + \arg w = \pi \Rightarrow zw = -|z|^2 = -z\bar{z} \Rightarrow z = -\bar{w}$.

**Grade** - Medium

Practice map, main concept: Sheet: J-I 4-5, 35; J-M 1, 22 | DPP: 1(7)

Also used in: Sheet: J-I 10; J-A 13 | DPP: 4(3); 6(2)

**Question 41**
How to convert look-alike forms of the polar form into the correct polar form?

**Model Answer**
$\cos\theta - i\sin\theta = e^{-i\theta}$; $\sin\theta + i\cos\theta = e^{i(\pi/2-\theta)}$; $-\cos\theta - i\sin\theta = e^{i(\pi+\theta)}$; $1+\cos\theta+i\sin\theta = 2\cos\dfrac{\theta}{2}\,e^{i\theta/2}$; $1 + i\tan\theta = \sec\theta\, e^{i\theta}$.

Check the sign of the modulus: $1+\cos\dfrac{10\pi}{9}+i\sin\dfrac{10\pi}{9} = 2\cos\dfrac{5\pi}{9}e^{i5\pi/9} = -2\cos\dfrac{5\pi}{9}\,e^{-i4\pi/9}$, so $|z| = -2\cos\dfrac{5\pi}{9}$, $\arg z = -\dfrac{4\pi}{9}$.

**Grade** - Low
Practice map, main concept: Sheet: J-II 17, 42; C-I 6 | DPP: 2(19-20)
Also used in: Sheet: J-I 1 | DPP: 3(23)

**Question 42**
When $|z| = 1$, what to do when $(1+z)$ or $(1-z)$ is used in questions?

**Model Answer**

$$z = e^{i\theta}: \quad 1+z = 2\cos\frac{\theta}{2}\,e^{i\theta/2}, \qquad 1-z = -2i\sin\frac{\theta}{2}\,e^{i\theta/2}$$

Hence $\dfrac{1+z}{1+\bar{z}} = z$, $\arg\dfrac{1+z}{1+\bar{z}} = \theta$, and $\dfrac{1-z}{1+z} = -i\tan\dfrac{\theta}{2}$ (purely imaginary).

**Grade** - Low
Practice map, main concept: Sheet: J-I 10, 31; C-II 31-32
Also used in: DPP: 3(5)

## Geometrical Interpretation

**Question 43**
What is the distance formula for the complex plane?

**Model Answer**

$$|z_1 - z_2| = \sqrt{(x_1-x_2)^2 + (y_1-y_2)^2}$$

Ex: distance between $1+2i$ and $4-2i$ is $|-3+4i| = 5$.

**Grade** - Medium
Practice map, main concept: Sheet: C-II 24
Also used in: Sheet: J-II 14, 16; J-M 24, 39; J-A 18 | DPP: 3(4); 6(7)

**Question 44**
What does $z_2 - z_1$ represent on the complex plane (similar to a vector)?

**Model Answer**
$z_2 - z_1$ is the vector $\vec{PQ}$ from $P(z_1)$ to $Q(z_2)$: length $|z_2 - z_1|$, direction $\arg(z_2 - z_1)$.
Ex: $P(1+i)$, $Q(3+3i)$: $\vec{PQ} = 2+2i$, length $2\sqrt{2}$, direction $\dfrac{\pi}{4}$.

**Grade** - Low
Practice map, main concept: none
Also used in: none

**Question 45**

What does $|z_1 - z_2|$ represent on the complex plane?

**Model Answer**

The distance between the points $z_1$ and $z_2$. So $|z - z_0|$ with $z$ on a circle (centre $c$, radius $r$) has max $|z_0 - c| + r$ and min $\big||z_0 - c| - r\big|$.

Ex: $|z + 4| \le 3$: $|z + 1|$ has max $3 + 3 = 6$, min $0$ (as $-1$ lies inside the disc).

**Grade - High**

Practice map, main concept: Sheet: J-I 8, 23; J-II 30; J-M 15; J-A 5, 15, 20; C-I 12 | DPP: 2(10); 4(4) | Question Bank: 322, 342

Also used in: Sheet: J-I 25; J-II 29; J-M 6, 14, 16, 20, 29, 33, 35-36, 41; C-II 25 | DPP: 3(7) | Question Bank: 308, 330, 335-336, 356

**Question 46**

What is the expression of the locus which represents a circle?

**Model Answer**

$|z - z_0| = r$: circle; $|z - z_0| < r$: inside; $r_1 < |z - z_0| < r_2$: annulus (ring).

Ex: $|2z - 1| = 3 \iff \left|z - \frac{1}{2}\right| = \frac{3}{2}$: centre $\left(\frac{1}{2}, 0\right)$, radius $\frac{3}{2}$.

**Grade - High**

Practice map, main concept: Sheet: J-M 20, 31, 35 | DPP: 2(3); 3(11); 4(15)

Also used in: Sheet: J-I 8, 11, 23; J-II 20, 25-27, 30, 45; J-M 5-6, 15, 25, 28, 38, 44; J-A 6, 11, 17, 28; C-I 18, 22; C-II 15 | DPP: 2(10, 15); 3(22); 4(1, 4) | Question Bank: 317, 322, 324, 326-327, 345, 350, 352

**Question 47**

Relate all the locus results of straight line based on the distance formula to their application in complex numbers.

**Model Answer**

$|z-z_1| = |z-z_2|$: perpendicular bisector; $\left|\frac{z - z_1}{z - z_2}\right| = k \neq 1$: circle (Apollonius); $|z-z_1|+|z-z_2| = k$: ellipse if $k > |z_1 - z_2|$, segment if $k = |z_1 - z_2|$, empty if smaller.

$\big||z - z_1| - |z - z_2|\big| = k$: hyperbola if $k < |z_1 - z_2|$, two rays if equal. Ex: $|z - 4| + |z + 4| = 16$: ellipse $\frac{x^2}{64} + \frac{y^2}{48} = 1$.

**Grade - Medium**

Practice map, main concept: Sheet: J-I 21-22; J-II 22-23, 27, 29, 38, 51; J-M 25, 38, 44; J-A 28; C-I 4, 24 | DPP: 2(2, 11); 3(21-22); 5(2, 7) | Question Bank: 318, 331, 350

Also used in: Sheet: J-II 39-40, 49; J-M 23; C-II 8 | DPP: 6(6) | Question Bank: 348

**Question 48**

What is the section formula for internal and external division in the complex plane, and its similarity with vectors and straight line?

**Model Answer**

Internal: $z = \frac{mz_2 + nz_1}{m + n}$; external: $z = \frac{mz_2 - nz_1}{m - n}$.

Same as for position vectors / coordinates. Ex: midpoint of $2 + 4i$ and $6 - 2i$ is $4 + i$.

**Grade - Low**

Practice map, main concept: Sheet: J-II 44; C-II 1

Also used in: none

**Question 49**

What are the results for the centroid / incentre / excentres / orthocentre / circumcentre of a triangle?

**Model Answer**

$$G = \frac{z_1 + z_2 + z_3}{3}, \quad I = \frac{az_1 + bz_2 + cz_3}{a + b + c}, \quad I_1 = \frac{-az_1 + bz_2 + cz_3}{-a + b + c}$$

$$H = \frac{\sum z_1 \tan A}{\sum \tan A}, \qquad O = \frac{\sum z_1 \sin 2A}{\sum \sin 2A}$$

If the circumcentre is at the origin, orthocentre $= z_1 + z_2 + z_3$. Also $\sum |z - z_k|^2$ is least at $z = G$.

**Grade** - Medium

Practice map, main concept: Sheet: J-II 14; J-M 39; C-I 2 | DPP: 3(14)

Also used in: Sheet: J-II 24

**Question 50**

What do 0, $z_1$, $z_2$ and $z_1 + z_2$ represent on a complex plane, and what is the importance of $z_1 - z_2$ in this situation?

**Model Answer**

Vertices of a parallelogram; $z_1 + z_2$ and $z_1 - z_2$ are its diagonals. $|z_1 + z_2| = |z_1 - z_2| \Rightarrow$ rectangle ($\arg \frac{z_1}{z_2} = \pm\frac{\pi}{2}$); $|z_1| = |z_2| \Rightarrow$ rhombus.

Points $z_1, z_2, z_3, z_4$ (in order) form a parallelogram iff $z_1 + z_3 = z_2 + z_4$ (diagonals bisect).

**Grade** - Low

Practice map, main concept: Sheet: J-I 15 | DPP: 2(6); 3(18); 5(9)

Also used in: none

**Question 51**

How to find the equation of the locus of a point satisfying given conditions in a complex plane, by putting $z = x + iy$?

**Model Answer**

Put $z = x + iy$ ($\bar{z} = x - iy$), simplify the condition into a real equation in $x, y$, and identify the curve (line, circle, conic).

Ex: $|z| = \mathrm{Re}(z) + 2$: $\sqrt{x^2 + y^2} = x + 2 \Rightarrow y^2 = 4x + 4$ (parabola).

**Grade** - High

Practice map, main concept: Sheet: J-I 18, 20; J-II 26, 36-37, 39-40; J-M 5, 8, 17, 41; J-A 4, 11-12; C-I 17, 22; C-II 23 | DPP: 3(20) | Question Bank: 313, 328, 330

Also used in: Sheet: J-I 14, 16, 24; J-II 41, 51; J-M 12; J-A 5; C-II 24 | DPP: 3(10) | Question Bank: 338-339

**Question 52**

How to solve questions on the locus of a point satisfying given conditions in a complex plane, by relating with coordinate geometry?

**Model Answer**

Read each modulus as a distance and each argument as an angle; recognise the standard figure (circle, bisector, conic, arc) and use its geometry (tangent, perpendicular distance, extreme points) without expanding.

Ex: $|z_1| = 12$, $|z_2 - 3 - 4i| = 5$: second circle (centre at distance 5, radius 5) lies inside the first, so $\min |z_1 - z_2| = 12 - 10 = 2$.

**Grade** - Low

Practice map, main concept: Sheet: C-II 10

Also used in: none

**Question 53**
Explain the triangle inequality for complex numbers.

**Model Answer**

$$\big||z_1| - |z_2|\big| \le |z_1 \pm z_2| \le |z_1| + |z_2|$$

Ex: $|z| \ge 2 \Rightarrow \left|z + \frac{1}{2}\right| \ge |z| - \frac{1}{2} \ge \frac{3}{2}$, equality at $z = -2$.

**Grade** - Medium

Practice map, main concept: Sheet: J-I 25; J-II 53; J-M 43; J-A 17; C-I 5; C-II 4, 25 | DPP: 4(2) | Question Bank: 325, 336

Also used in: Sheet: C-I 4, 12 | DPP: 3(1)

**Question 54**
What is the situation if we have equality in the triangle inequality?

**Model Answer**

$|z_1 + z_2| = |z_1| + |z_2| \iff O, z_1, z_2$ collinear with $z_1, z_2$ on the same side of $O$ ($\arg z_1 = \arg z_2$);
$|z_1 + z_2| = \big||z_1| - |z_2|\big| \iff$ opposite sides ($\arg z_1 - \arg z_2 = \pi$).
Also $|z - z_1| + |z - z_2| = |z_1 - z_2| \iff z$ lies on the segment $z_1 z_2$.

**Grade** - Medium

Practice map, main concept: Sheet: J-M 36 | DPP: 3(1, 7); 5(5)

Also used in: Sheet: J-II 43, 47, 53

**Question 55**
Use of $z\bar{z} = |z|^2$ in questions.

**Model Answer**

Square a modulus as a product: $|z_1 \pm z_2|^2 = |z_1|^2 + |z_2|^2 \pm 2\,\mathrm{Re}(z_1\bar{z}_2)$; this proves identities and turns $|z| = r$ into $\bar{z} = \frac{r^2}{z}$.

Ex: $|z_1 + z_2|^2 + |z_1 - z_2|^2 = 2\left(|z_1|^2 + |z_2|^2\right)$; $|1 - z_1\bar{z}_2|^2 - |z_1 - z_2|^2 = (1 - |z_1|^2)(1 - |z_2|^2)$.

**Grade** - Medium

Practice map, main concept: Sheet: J-I 11, 14; J-II 3-4, 19; J-A 6; C-I 9, 20 | DPP: 3(3); 6(7) | Question Bank: 327, 352

Also used in: Sheet: J-I 17; J-II 34; J-A 16, 21; C-I 19; C-II 18, 22 | DPP: 6(15) | Question Bank: 325

## Angles

**Question 56**
Relate $\arg(z)$ with a vector.

**Model Answer**

$\arg z$ is the direction of the vector $\vec{Oz}$; unit vector along it is $\frac{z}{|z|}$. Sum of unit vectors gives the angle bisector.

Ex: bisector of the angle between $3 + 4i$ and $5i$ has direction $\frac{3+4i}{5} + i = \frac{3+9i}{5}$.

**Grade** - Low

Practice map, main concept: DPP: 4(7)

Also used in: none

**Question 57**

How to write any angle of a line on the complex plane as $\arg(z_2 - z_1)$?

**Model Answer**

The line from $A(z_1)$ to $B(z_2)$ makes angle $\arg(z_2 - z_1)$ with the positive real axis. $\arg(z - z_1) = \theta$ is a ray from $z_1$ (excluding $z_1$).

Ex: $\arg(z - 2 - 3i) = \dfrac{\pi}{4}$: ray $y = x + 1$, $x > 2$.

**Grade** - Medium

Practice map, main concept: Sheet: C-I 13 | DPP: 2(14)

Also used in: Sheet: J-M 17

**Question 58**

How to write any angle between two lines on the complex plane as $\arg\left(\dfrac{z_3 - z_1}{z_2 - z_1}\right)$?

**Model Answer**

Angle from $\vec{AB}$ to $\vec{AC}$ (anticlockwise) $= \arg(z_3 - z_1) - \arg(z_2 - z_1) = \arg\dfrac{z_3 - z_1}{z_2 - z_1}$.

Ex: $A = 0$, $B = 1$, $C = 1 + i$: $\angle BAC = \arg(1 + i) = \dfrac{\pi}{4}$.

**Grade** - Medium

Practice map, main concept: none

Also used in: Sheet: J-A 2 | DPP: 5(4)

**Question 59**

What is the use of the initial and final vector in writing an angle?

**Model Answer**

$\arg\left(\dfrac{\text{final vector}}{\text{initial vector}}\right)$ = angle turned anticlockwise from the initial to the final vector.

Both vectors must start from the vertex of the angle. Ex: angle at $z_1$ from $z_1\vec{z}_2$ to $z_1\vec{z}_3$ is $\arg\dfrac{z_3 - z_1}{z_2 - z_1}$.

**Grade** - Low

Practice map, main concept: none

Also used in: none

**Question 60**

If the angle is measured in the clockwise direction, then what to do?

**Model Answer**

Take it negative: a clockwise angle $\theta$ is $-\theta$, i.e. $\arg\dfrac{\text{final}}{\text{initial}} = -\theta$ (or interchange initial and final).

Ex: rotating $z_1 = 3 + 5i$ clockwise by $\dfrac{\pi}{2}$ gives $z_1 e^{-i\pi/2} = 5 - 3i$.

**Grade** - Low

Practice map, main concept: none

Also used in: none

**Question 61**
What to keep in mind while marking the angle between two line segments?

**Model Answer**
Both segments must point away from the common vertex, and the sense (anticlockwise +, clockwise −) must be fixed; otherwise the angle becomes $\pi - \theta$ or $-\theta$.

Ex: $\arg\dfrac{z_1 - z}{z_2 - z}$ is the angle at $z$, not at $z_1$.

**Grade** - Low

Practice map, main concept: none

Also used in: none

**Question 62**
What is the condition for two lines to be parallel?

**Model Answer**
$AB \parallel CD \iff \dfrac{z_4 - z_3}{z_2 - z_1}$ is purely real.

Collinear points: $\dfrac{z_3 - z_1}{z_2 - z_1}$ real. Ex: $1 + i,\ 2 + 2i,\ 4 + 4i$ are collinear (ratio 3).

**Grade** - Low

Practice map, main concept: none

Also used in: none

**Question 63**
What is the condition for two lines to be perpendicular?

**Model Answer**
$AB \perp CD \iff \dfrac{z_4 - z_3}{z_2 - z_1}$ is purely imaginary.

Ex: $|z_1 + z_2|^2 = |z_1|^2 + |z_2|^2 \Rightarrow \mathrm{Re}(z_1\bar{z}_2) = 0 \Rightarrow \dfrac{z_1}{z_2}$ purely imaginary ($Oz_1 \perp Oz_2$).

**Grade** - Medium

Practice map, main concept: Sheet: J-II 24; J-M 21

Also used in: Sheet: J-II 50; C-I 20 | DPP: 2(6); 3(18); 5(9)

**Question 64**
Application of the locus of $P$ when the angle between $PA$ and $PB$ is fixed, in complex numbers.

**Model Answer**
$\arg\dfrac{z - z_1}{z - z_2} = \theta$ $(0 < \theta < \pi)$: arc of a circle through $z_1, z_2$ (endpoints excluded); $\theta = \pm\dfrac{\pi}{2}$: semicircle on diameter $z_1z_2$; $\theta = 0$ or $\pi$: parts of the line $z_1z_2$.

Ex: $\arg\dfrac{z - 1}{z + 1} = \dfrac{\pi}{4}$: chord of length 2 subtends 90° at the centre: centre $(0, 1)$, radius $\sqrt{2}$ (upper arc).

**Grade** - Medium

Practice map, main concept: Sheet: J-M 10, 16, 23, 28; J-A 19 | DPP: 5(8)

Also used in: Sheet: J-II 23, 52; J-M 14

## Rotation

### Question 65

How to use the result of rotation when two line segments are given?

**Model Answer**

$$\frac{z_3 - z_1}{z_2 - z_1} = \frac{|z_3 - z_1|}{|z_2 - z_1|} e^{i\theta}$$

Here $\theta = \angle BAC$ measured anticlockwise from $AB$ to $AC$.

Ex: square $ABCD$ (anticlockwise), $A = z_1$, $B = z_2$: $D = z_1 + i(z_2 - z_1)$, $C = z_2 + i(z_2 - z_1)$.

**Grade** - **Medium**

Practice map, main concept: Sheet: J-I 27, 32, 34; J-M 37; C-II 12, 15 | DPP: 3(9); 4(6, 8-9) | Question Bank: 321

Also used in: Sheet: J-II 21, 28; C-II 10

### Question 66

What happens to a vector in the complex plane if we multiply it by $e^{i\theta}$?

**Model Answer**

It rotates through $\theta$ anticlockwise about the origin, length unchanged ($|e^{i\theta}| = 1$). Multiplying by $ke^{i\theta}$ also stretches it $k$ times.

Ex: $-4 + 5i$ turned through $180°$ and stretched 1.5 times: $1.5(-1)(-4 + 5i) = 6 - \frac{15}{2}i$.

**Grade** - **Medium**

Practice map, main concept: Sheet: J-I 30, 33

Also used in: Question Bank: 321

### Question 67

What happens to a vector in the complex plane if we multiply it by $i$?

**Model Answer**

$i = e^{i\pi/2}$: rotation through $90°$ anticlockwise; $-i$: $90°$ clockwise.

Ex: $i(5 + 4i) = -4 + 5i$.

**Grade** - **Medium**

Practice map, main concept: Question Bank: 340

Also used in: Sheet: J-I 27; J-M 37; C-I 3; C-II 12 | DPP: 3(9)

### Question 68

What is the relation between $z_1$, $z_2$ and $z_3$ if these are vertices of an equilateral triangle?

**Model Answer**

$$z_1^2 + z_2^2 + z_3^2 = z_1z_2 + z_2z_3 + z_3z_1 \iff \frac{1}{z_1 - z_2} + \frac{1}{z_2 - z_3} + \frac{1}{z_3 - z_1} = 0$$

Equivalently $z_1 + \omega z_2 + \omega^2 z_3 = 0$ (or with $\omega, \omega^2$ swapped); centroid = circumcentre $z_0$ gives $z_1^2 + z_2^2 + z_3^2 = 3z_0^2$. Roots of $z^3 - 3\alpha z^2 + 3\beta z + \gamma = 0$ form one iff $\alpha^2 = \beta$.

**Grade** - **Medium**

Practice map, main concept: Sheet: J-II 21, 46; C-I 25 | DPP: 4(5); 5(10) | Question Bank: 310

Also used in: Sheet: J-I 28, 32

**Question 69**
How to find the remaining vertices of a regular polygon of $n$ sides if one vertex and its circumcentre are known?

**Model Answer**

$$z_k = z_0 + (z_1 - z_0)\, e^{i2\pi(k-1)/n}, \qquad k = 1, 2, \ldots, n$$

Ex: equilateral triangle in $|z| = 2$ with $z_1 = 1 + i\sqrt{3}$: $z_2 = z_1 e^{2\pi i/3} = -2$, $z_3 = 1 - i\sqrt{3}$.

**Grade** - Low
Practice map, main concept: Sheet: J-I 28-29; J-II 28
Also used in: Sheet: J-I 34

## De-Moivre's Theorem (DMT)

**Question 70**
How to find an integral power $z^n$ (or $\alpha^n + \beta^n$ for complex roots) using DMT?

**Model Answer**

$$(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta, \qquad (re^{i\theta})^n = r^n e^{in\theta}\ (n \in \mathbb{Z})$$

Write each number (or each root of the quadratic) in exponential form, raise the power, then reduce $n\theta$ by multiples of $2\pi$.

Ex: roots of $x^2 - \sqrt{2}x + 2 = 0$ are $\sqrt{2}e^{\pm i\pi/3}$, so $\alpha^{14} + \beta^{14} = 2 \cdot 2^7 \cos\dfrac{14\pi}{3} = -128$; and $(\sqrt{3}+i)^{100} = 2^{100}e^{i50\pi/3} = 2^{99}(-1 + i\sqrt{3})$.

**Grade** - Medium
Practice map, main concept: Sheet: J-I 37; J-M 2, 11, 18, 42; C-I 15-16 | DPP: 2(12); 5(1); 6(9) | Question Bank: 343, 354
Also used in: Sheet: J-I 13; J-II 13, 17, 31, 42; C-I 6, 21 | Question Bank: 320

**Question 71**
What are the steps to find the $n^{\text{th}}$ root of a complex number using DMT?

**Model Answer**
(i) Write $z = re^{i\theta}$; (ii) replace $\theta$ by $\theta + 2k\pi$; (iii) $z^{1/n} = r^{1/n}e^{i(\theta+2k\pi)/n}$; (iv) put $k = 0, 1, \ldots, n-1$ ($n$ distinct roots on the circle of radius $r^{1/n}$).
Ex: $\omega^3 = 8i = 8e^{i\pi/2}$: $\omega = 2e^{i(\pi/6+2k\pi/3)}$; second-quadrant root ($k = 1$): $2e^{i5\pi/6} = -\sqrt{3} + i$.

**Grade** - Low
Practice map, main concept: Sheet: J-I 36, 38; J-II 48; C-II 8 | DPP: 5(3, 6); 6(1, 6, 10)
Also used in: Sheet: C-I 24; C-II 5

**Question 72**
Why do we use the general value of the argument in place of the argument while using DMT for the $n^{\text{th}}$ root?

**Model Answer**
$e^{i\theta} = e^{i(\theta+2k\pi)}$, but after dividing by $n$ the angles $\dfrac{\theta + 2k\pi}{n}$ give different points; with only $\theta$ we would get just one of the $n$ roots.
Ex: $1^{1/3}$ with $\theta = 0$ gives only 1; with $2k\pi$ we get $1, \omega, \omega^2$.

**Grade** - Low
Practice map, main concept: none
Also used in: Sheet: J-I 38

**Question 73**
How to solve questions where it is given that an equation has a purely real / purely imaginary solution?

**Model Answer**
Put $z = x$ (real) or $z = iy$; separate real and imaginary parts; both must vanish, so the real $x$ (or $y$) is a common root of two real equations.
Ex: $z^2 - (3+i)z + m + 2i = 0$ has a real root $x$: $x^2 - 3x + m = 0$ and $-x + 2 = 0 \Rightarrow x = 2,\ m = 2$.

**Grade** - Low
Practice map, main concept: Sheet: J-II 7, 15-16; C-I 11 | DPP: 2(9); 3(6)
Also used in: none

## Cube Roots of Unity

**Question 74**
What are the three cube roots of unity and how are they represented?

**Model Answer**

$$z^3 = 1:\ 1,\ \omega = \frac{-1+i\sqrt{3}}{2} = e^{i2\pi/3},\ \omega^2 = \frac{-1-i\sqrt{3}}{2} = e^{-i2\pi/3} = \bar{\omega}$$

Ex: $\omega \cdot \omega^2 = \omega^3 = 1$ and $|\omega| = 1$.

**Grade** - Medium
Practice map, main concept: Sheet: J-I 40
Also used in: Sheet: J-I 19, 42; J-M 26

**Question 75**
What is the sum of all three cube roots of unity?

**Model Answer**

$$1 + \omega + \omega^2 = 0$$

So $1 + \omega = -\omega^2$, $1 + \omega^2 = -\omega$. Ex: $(1 + \omega - \omega^2)(1 + \omega^2 - \omega) = (-2\omega^2)(-2\omega) = 4$.

**Grade** - Medium
Practice map, main concept: Sheet: J-I 39
Also used in: Sheet: J-I 41; C-II 29-30 | DPP: 4(13); 6(12) | Question Bank: 351

**Question 76**
How to find higher integral powers of $\omega$?

**Model Answer**
Divide the power by 3: $\omega^{3k} = 1$, $\omega^{3k+1} = \omega$, $\omega^{3k+2} = \omega^2$.
Ex: $(1+\omega)^7 = (-\omega^2)^7 = -\omega^{14} = -\omega^2 = 1 + \omega$, so $A = B = 1$.

**Grade** - High
Practice map, main concept: Sheet: J-I 41-42; J-II 33, 52; J-M 19, 30, 45; J-A 8, 10, 29; C-II 3, 29-30 | DPP: 4(10, 13-14); 5(14); 6(12, 14) | Question Bank: 314, 333, 348, 351
Also used in: Sheet: J-I 39, 43; J-M 27, 47; J-A 3 | DPP: 4(11-12); 6(13) | Question Bank: 310, 346

**Question 77**
How to find the sum of any $r^{\text{th}}$ power of all cube roots of unity?

**Model Answer**
$1+\omega^r+\omega^{2r}=3$ if $r$ is a multiple of 3, otherwise 0.
Ex: $\omega^n+\omega^{2n}=-1$ for $n=3k+2$.

**Grade** - Medium
Practice map, main concept: Sheet: J-A 3 | DPP: 4(12); 6(13) | Question Bank: 346
Also used in: Sheet: J-M 30; J-A 10

**Question 78**
What do all three cube roots of unity represent on the complex plane, and can you find the dimensions of that triangle?

**Model Answer**
Vertices of an equilateral triangle inscribed in $|z|=1$ (centroid at $O$): side $\sqrt{3}$, area $\dfrac{3\sqrt{3}}{4}$.
Ex: $|1-\omega|=\left|\dfrac{3}{2}-\dfrac{\sqrt{3}}{2}i\right|=\sqrt{3}$.

**Grade** - Low
Practice map, main concept: none
Also used in: Sheet: J-II 46

**Question 79**
What happens to a vector in the complex plane if we multiply it by $\omega$?

**Model Answer**
$\omega=e^{i2\pi/3}$: rotation through 120° anticlockwise (by $\omega^2$: 120° clockwise), length unchanged.
Ex: $\dfrac{b+c\omega+a\omega^2}{a+b\omega+c\omega^2}=\omega^2$ since $\omega^2(a+b\omega+c\omega^2)=a\omega^2+b+c\omega$.

**Grade** - Low
Practice map, main concept: DPP: 4(11)
Also used in: DPP: 4(6)

**Question 80**
Write all factors which are to be remembered.

**Model Answer**

$$x^2+x+1=(x-\omega)(x-\omega^2),\qquad x^2-x+1=(x+\omega)(x+\omega^2)$$
$$a^3+b^3+c^3-3abc=(a+b+c)(a+b\omega+c\omega^2)(a+b\omega^2+c\omega)$$
$$a^2+b^2+c^2-ab-bc-ca=(a+b\omega+c\omega^2)(a+b\omega^2+c\omega)=|a+b\omega+c\omega^2|^2\ (a,b,c\in\mathbb{R})$$

Ex: distinct integers $a,b,c$: $\min|a+b\omega+c\omega^2|^2=\frac{1}{2}[(a-b)^2+(b-c)^2+(c-a)^2]_{\min}=\frac{1}{2}(1+1+4)=3$.

**Grade** - Medium
Practice map, main concept: Sheet: J-II 32; J-A 16; C-II 22 | DPP: 6(15)
Also used in: Sheet: J-II 33; C-II 11 | Question Bank: 333

**Question 81**
Write $-\omega$ and $-\omega^2$ in exponential form.

**Model Answer**

$$-\omega = e^{-i\pi/3} = \frac{1 - i\sqrt{3}}{2}, \qquad -\omega^2 = e^{i\pi/3} = \frac{1 + i\sqrt{3}}{2}$$

These are the sixth roots of unity; $(-\omega)^6 = 1$. Ex: $(1 + i\sqrt{3}) = -2\omega^2$, so $(1 + i\sqrt{3})^p + (1 - i\sqrt{3})^p = (-2)^p(\omega^{2p} + \omega^p)$.

**Grade** - Medium
Practice map, main concept: none
Also used in: Sheet: J-I 40; J-M 19; J-A 29; C-II 3 | DPP: 4(10); 6(14)

**Question 82**
What are the roots of the equations $x^2 + x + 1 = 0$ and $x^2 - x + 1 = 0$?

**Model Answer**

$$x^2 + x + 1 = 0 : \ x = \omega, \omega^2; \qquad x^2 - x + 1 = 0 : \ x = -\omega, -\omega^2$$

Ex: roots of $x^2 - x + 1 = 0$: $\alpha^{101} + \beta^{107} = (-\omega)^{101} + (-\omega^2)^{107} = -\omega^2 - \omega = 1$.

**Grade** - Medium
Practice map, main concept: Sheet: J-I 43; J-M 27
Also used in: Sheet: J-M 2; C-II 17 | DPP: 5(6, 10)

## $n^{\text{th}}$ Roots of Unity

**Question 83**
What are the $n^{\text{th}}$ roots of unity and how are they represented?

**Model Answer**

$$z^n = 1 : \ z = e^{i2k\pi/n}, \ k = 0, 1, \ldots, n-1; \ \text{i.e. } 1, \alpha, \alpha^2, \ldots, \alpha^{n-1}, \ \alpha = e^{i2\pi/n}$$

Ex: fourth roots: $1, i, -1, -i$.

**Grade** - Medium
Practice map, main concept: Sheet: J-M 13; J-A 2 | DPP: 1(8)
Also used in: Sheet: J-M 34

**Question 84**
What is the sum of all $n^{\text{th}}$ roots of unity?

**Model Answer**

$$1 + \alpha + \alpha^2 + \cdots + \alpha^{n-1} = 0 \quad (n \geq 2)$$

Ex: $\sum_{k=1}^{6} \left( \cos \frac{2k\pi}{7} + i \sin \frac{2k\pi}{7} \right) = -1$.

**Grade** - Low
Practice map, main concept: none
Also used in: Sheet: J-I 44; C-II 16

**Question 85**
What is the product of all $n^{\text{th}}$ roots of unity?

**Model Answer**

$$1 \cdot \alpha \cdot \alpha^2 \cdots \alpha^{n-1} = (-1)^{n-1}$$

Ex: $n = 4$: $1 \cdot i \cdot (-1) \cdot (-i) = -1 = (-1)^3$.

**Grade** - Low
Practice map, main concept: none
Also used in: DPP: 5(13)

**Question 86**
What is the common ratio of the GP formed by all $n^{\text{th}}$ roots of unity?

**Model Answer**
Common ratio $\alpha = e^{i2\pi/n}$. Common roots of $1 + z + \cdots + z^{17} = 0$ and $1 + z + \cdots + z^{13} = 0$: $z^{18} = 1 = z^{14}$, $z \neq 1$, so $z^2 = 1$, $z = -1$ only.

**Grade** - Medium
Practice map, main concept: none
Also used in: Sheet: J-M 13 | DPP: 1(8)

**Question 87**
How to find the sum of any $r^{\text{th}}$ power of all $n^{\text{th}}$ roots of unity?

**Model Answer**

$$\sum_{k=0}^{n-1} \alpha^{kr} = \begin{cases} n, & n \mid r \\ 0, & n \nmid r \end{cases}$$

Ex: eighth roots: $\sum \alpha_k^3 = 0$, $\sum \alpha_k^8 = 8$.

**Grade** - Medium
Practice map, main concept: Sheet: J-I 44; J-M 34; C-II 16 | DPP: 5(11-13); 6(11)
Also used in: DPP: 6(5)

**Question 88**
What do all $n^{\text{th}}$ roots of unity represent on the complex plane, and can you find the dimensions of that polygon?

**Model Answer**
Vertices of a regular $n$-gon inscribed in $|z| = 1$: side $2\sin\dfrac{\pi}{n}$, area $\dfrac{n}{2}\sin\dfrac{2\pi}{n}$.

Ex: roots of $z^6 = 1$ form a regular hexagon of side 1 and area $\dfrac{3\sqrt{3}}{2}$.

**Grade** - Medium
Practice map, main concept: Sheet: J-A 9; C-II 19
Also used in: DPP: 5(3)

**Question 89**
What are the factors of $(x^n - 1)$ involving $n^{\text{th}}$ roots of unity?

**Model Answer**

$$x^n - 1 = (x-1)(x-\alpha)(x-\alpha^2)\cdots(x-\alpha^{n-1}), \qquad \prod_{k=1}^{n-1}(x-\alpha^k) = 1 + x + \cdots + x^{n-1}$$

Ex: $\prod_{k=1}^{n-1}(2-\alpha^k) = 2^n - 1$; $\prod_{k=1}^{n-1}|1-\alpha^k| = n$.

**Grade** - Medium

Practice map, main concept: Sheet: J-A 7, 26; C-I 23, 26 | DPP: 6(5)
Also used in: Sheet: J-II 35

**Question 90**
What are the factors of $(x^n - 1)$ without involving $n^{\text{th}}$ roots of unity?

**Model Answer**

Pair each root with its conjugate: $(x-\alpha^k)(x-\alpha^{n-k}) = x^2 - 2x\cos\dfrac{2k\pi}{n} + 1$ (real quadratic factors).

Ex: $z^5 - 32 = (z-2)\left(z^2 - 4z\cos\dfrac{2\pi}{5} + 4\right)\left(z^2 - 4z\cos\dfrac{4\pi}{5} + 4\right)$.

**Grade** - Low

Practice map, main concept: Sheet: J-II 35; C-II 9
Also used in: none

**Question 91**
What is the modulus of any $n^{\text{th}}$ root of unity?

**Model Answer**

$|\alpha^k| = 1$; all lie on the unit circle, so $\bar{\alpha}^k = \dfrac{1}{\alpha^k}$. Distance between consecutive roots $= 2\sin\dfrac{\pi}{n}$.

Ex: $|\alpha_{k+1} - \alpha_k| = 2\sin\dfrac{\pi}{14}$ for the 14th roots $\alpha_k = e^{ik\pi/7}$.

**Grade** - Medium

Practice map, main concept: none
Also used in: Sheet: J-A 9, 26

**Question 92**
Which two $n^{\text{th}}$ roots of unity are conjugates of each other?

**Model Answer**

$$\overline{\alpha^k} = \alpha^{n-k}$$

Ex: $n = 5$: $\bar{\alpha} = \alpha^4$, $\overline{\alpha^2} = \alpha^3$.

**Grade** - Medium

Practice map, main concept: none
Also used in: Sheet: J-A 7 | DPP: 5(11)

**Question 93**
The product of which two $n^{\text{th}}$ roots of unity is 1?

**Model Answer**

$$\alpha^k \cdot \alpha^{n-k} = \alpha^n = 1$$

Ex: $\alpha^5 = 1$, $\alpha \neq 1$: $1 + \alpha + \alpha^2 + \alpha^{-2} - \alpha^{-1} = 1 + \alpha + \alpha^2 + \alpha^3 - \alpha^4 = -2\alpha^4$, modulus 2.

**Grade** - Medium

Practice map, main concept: none

Also used in: Sheet: J-A 7 | DPP: 6(11)

## Summation of Series using Complex Numbers

**Question 94**
Prove the result of summation of a series of the type $\sum$(trig ratio) using complex numbers.

**Model Answer**
Form $C + iS = \sum e^{i(\alpha+k\beta)}$ (a GP), sum it, and take the real / imaginary part:

$$\sum_{k=0}^{n-1} \cos(\alpha + k\beta) = \frac{\sin\frac{n\beta}{2}}{\sin\frac{\beta}{2}} \cos\left(\alpha + \frac{(n-1)\beta}{2}\right)$$

Ex: $\sum_{m=1}^{15} \sin(2m-1)2^\circ = \dfrac{\sin^2 30^\circ}{\sin 2^\circ} = \dfrac{1}{4\sin 2^\circ}$.

**Grade** - Low

Practice map, main concept: Sheet: J-I 13; J-II 34

Also used in: Sheet: C-II 13

## Complex Numbers and Binomial Coefficients

**Question 95**
How to solve questions of series based on binomial coefficients, and when to put $i$, when to put $\omega$?

**Model Answer**
In $(1+x)^n = \sum C_r x^r$ put $x = i$ when the terms change sign in pairs / skip by 2 with alternate signs, $x = \omega$ when suffixes skip by 3; then compare real and imaginary parts.

$$C_0 - C_2 + C_4 - \cdots = 2^{n/2}\cos\frac{n\pi}{4}, \qquad C_1 - C_3 + C_5 - \cdots = 2^{n/2}\sin\frac{n\pi}{4}$$

**Grade** - Medium

Practice map, main concept: Sheet: C-II 13 | Question Bank: 347

Also used in: none

**Question 96**
How to find the sum of coefficients of all powers of the form $x^{3m}$, $x^{3m+1}$ and $x^{3m+2}$ in any expansion?

**Model Answer**

$$\sum a_{3m} = \frac{f(1) + f(\omega) + f(\omega^2)}{3}, \quad \sum a_{3m+1} = \frac{f(1) + \omega^2 f(\omega) + \omega f(\omega^2)}{3}, \quad \sum a_{3m+2} = \frac{f(1) + \omega f(\omega) + \omega^2 f(\omega^2}{3}$$

Ex: $C_0 + C_3 + C_6 + \cdots = \dfrac{1}{3}\left(2^n + 2\cos\dfrac{n\pi}{3}\right)$; for $n = 6$: $1 + 20 + 1 = 22$.

**Grade** - Low

Practice map, main concept: Sheet: C-II 11, 26
Also used in: none

### Question 97

How to find the sum of coefficients of all powers of the form $x^{pm}$, $x^{pm+1}$, ..., $x^{pm+(p-1)}$ in any expansion?

**Model Answer**

$$\sum_m a_{pm+r} = \frac{1}{p}\sum_{k=0}^{p-1} \alpha^{-kr} f(\alpha^k), \qquad \alpha = e^{i2\pi/p}$$

Ex ($p=4$): $C_1 + C_5 + C_9 + \cdots = \frac{1}{4}\left(2^n + 2^{n/2+1}\sin\frac{n\pi}{4}\right)$; for $n=5$: $5+1=6$.

**Grade** - Low
Practice map, main concept: Sheet: C-II 27-28
Also used in: none

## Straight Line in Complex Numbers

### Question 98

How to assume any complex number on the line joining two given points?

**Model Answer**

$z = z_1 + t(z_2 - z_1) = (1-t)z_1 + tz_2$, $t \in \mathbb{R}$ ($0 \le t \le 1$ for the segment).

Equivalently $\dfrac{z - z_1}{z_2 - z_1}$ is real. Ex: on the segment from $-3+5i$ to $-5-3i$ the argument of $z$ varies from $\pi - \tan^{-1}\dfrac{5}{3}$ to $-\pi + \tan^{-1}\dfrac{3}{5}$ through $\pi$.

**Grade** - Medium
Practice map, main concept: Sheet: J-II 43, 47 | DPP: 3(13)
Also used in: DPP: 5(8) | Question Bank: 342

### Question 99

What is the general form of the equation of a line in the complex plane, and how to convert it into the form $ax + by + c = 0$?

**Model Answer**

$\bar{a}z + a\bar{z} + b = 0$ ($b \in \mathbb{R}$, $a \neq 0$); with $a = p + iq$: $2(px + qy) + b = 0$, slope $-\dfrac{p}{q}$.

Through $z_1, z_2$: $\begin{vmatrix} z & \bar{z} & 1 \\ z_1 & \bar{z}_1 & 1 \\ z_2 & \bar{z}_2 & 1 \end{vmatrix} = 0$. Ex: $(1+2i)z + (2i-1)\bar{z} = 10i$ with $z = x+iy$ gives $i(4x+2y) = 10i$, i.e. $2x + y = 5$ (imaginary-axis intercept 5).

**Grade** - High
Practice map, main concept: Sheet: J-II 25, 50; J-M 6, 29, 33; J-A 14; C-I 7, 27 | DPP: 6(3) | Question Bank: 317, 324, 326, 335, 341, 345, 356
Also used in: Sheet: J-A 4; C-II 23

**Question 100**

How to find the image of a point when the equation of the line is given in general form?

**Model Answer**

Image $z_2$ of $z_1$ in $\bar{a}z + a\bar{z} + b = 0$: $\bar{a}z_2 + a\bar{z}_1 + b = 0$, so $z_2 = -\dfrac{a\bar{z}_1 + b}{\bar{a}}$.

Ex: image of $z_1 = 2 + i$ in $z + \bar{z} = 0$ (imaginary axis, $a = 1, b = 0$): $z_2 = -\bar{z}_1 = -2 + i$.

**Grade** - Low

Practice map, main concept: none

Also used in: none

## Circle in Complex Numbers

**Question 101**

What is the general form of the equation of a circle in the complex plane, and how to convert it into the general form studied in the chapter Circle?

**Model Answer**

$z\bar{z} + \bar{a}z + a\bar{z} + b = 0$ ($b \in \mathbb{R}$): centre $-a$, radius $\sqrt{|a|^2 - b}$.

Put $z = x+iy$, $a = p+iq$: $x^2+y^2+2px+2qy+b = 0$. Diametric form: $|z-z_1|^2+|z-z_2|^2 = |z_1-z_2|^2$.

Ex: $z\bar{z} + (4 - 3i)z + (4 + 3i)\bar{z} + 5 = 0$: $a = 4 + 3i$, radius $\sqrt{25 - 5} = 2\sqrt{5}$.

**Grade** - Medium

Practice map, main concept: Sheet: J-I 19; J-II 49; C-II 18 | DPP: 3(12); 6(4)

Also used in: Sheet: J-A 19; C-I 7, 27 | Question Bank: 313

**Question 102**

How to check whether 4 given points on the complex plane are concyclic?

**Model Answer**

$z_1, z_2, z_3, z_4$ (in order) are concyclic $\iff \dfrac{(z_4 - z_1)(z_2 - z_3)}{(z_2 - z_1)(z_4 - z_3)}$ is purely real.

Ex: $1, i, -1, -i$: $\dfrac{(-i - 1)(i + 1)}{(i - 1)(-i + 1)} = \dfrac{-2i}{2i} = -1$ (real), so concyclic.

**Grade** - Medium

Practice map, main concept: none

Also used in: Sheet: J-A 13